

AI TRENDS

EXECUTIVE SUMMARY

Prepared for: Technical Leaders

Report Date: May 31, 2026

Research Window: March – May 2026 (Most Recent 3 Months)

Focus Areas: Generative AI | AI Infrastructure | AI Agents | Foundation Models

Report Focus: Balanced | Output Format: PDF

REPORT HIGHLIGHTS

- ▶ \$725B–\$830B in hyperscaler CapEx planned for 2026 — a 79% YoY increase
 - ▶ Anthropic raises \$65B at \$965B valuation, eclipsing OpenAI (May 28, 2026)
 - ▶ 85% of enterprises plan to customize AI agents for their workflows
 - ▶ GPT-5.5, Claude Opus 4.8, Gemini 3.5 all shipped in the past 3 months
 - ▶ Only 21% of companies have mature AI agent governance models
 - ▶ AI servers projected to surpass general-purpose servers in power consumption
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Sources: Gartner, Deloitte, Bloomberg, TrendForce, Stanford HAI, Anthropic, OpenAI, Google, Federal Reserve Atlanta, Harvard Business Review, International AI Safety Report 2026

1. EXECUTIVE OVERVIEW



Key Takeaways for Technical Leaders

- Agentic inflection point:** AI agents are no longer experimental. 85% of enterprises plan to customize agents for specific workflows (Deloitte, Jan 2026), and software engineering accounts for ~50% of AI agent API usage (Anthropic, 2026). Technical leaders must architect for agent orchestration, not just model inference.
- Unprecedented infrastructure spend:** The top four US hyperscalers have raised combined 2026 CapEx guidance to \$725B — primarily for AI data centers. The top nine global CSPs plan \$830B, a 79% YoY increase (TrendForce, May 2026). Gartner forecasts \$2.59T in total worldwide AI spending in 2026.
- Foundation model competition intensified:** GPT-5.5, Claude Opus 4.8, Gemini 3.5, and specialized models shipped in the past three months alone. Benchmark leaders: GPT-5.4-Pro at 94.4% on GPQA Diamond; Claude Opus 4.7 at 87.6% on SWE-Bench Verified (AI Release Tracker, May 2026).
- Valuation milestone:** Anthropic raised \$65B at a \$965B valuation on May 28, 2026 — surpassing OpenAI's \$852B post-money valuation. Both companies are preparing for IPOs, signaling a new phase of AI commercialization.
- Governance gap is critical:** Only 21% of companies deploying agentic AI have mature governance models (Deloitte, Jan 2026). 68% of organizations have experienced data leakage from employees sharing sensitive information with AI tools. Governance infrastructure must be built proactively.
- Energy as strategic constraint:** AI servers are projected to surpass general-purpose servers in total electricity consumption in 2026. Data center power capacity expected to reach ~155 GW globally (+29% YoY). Demand projected to double by 2030 (Gartner).

Report Scope & Methodology

Time window: March-May 2026 (most recent 3 months) | Audience: Technical Leaders | Focus: Balanced

Sources: Live internet research from Gartner, Deloitte, Bloomberg, TrendForce, Stanford HAI, official lab announcements, Federal Reserve Atlanta, Harvard Business Review, International AI Safety Report 2026, and industry analysts.

Note: No pre-existing workflow documents were available in the workspace; all content is sourced from live research.

2. AI LANDSCAPE SNAPSHOT

The AI landscape in mid-2026 is defined by a decisive phase transition: the industry has moved from experimentation to execution at scale. Three structural shifts characterize this moment:

From models to systems. The dominant paradigm has shifted from single-model inference to multi-agent systems that plan, execute, and iterate autonomously. Frameworks like LangGraph, CrewAI, Microsoft Semantic Kernel, and AutoGen have matured into production-grade tools.

From pilots to production — unevenly. Only 25% of organizations have moved 40%+ of AI pilots into production (Deloitte 2026), but 54% expect to reach that threshold within 3-6 months. The pilot-to-production gap remains the primary operational challenge.

From US-centric to globally contested. AI sovereignty has emerged as a central policy principle. 77% of companies now factor country of origin into AI vendor selection. Europe expanded state-backed AI supercomputing clusters from 3 to 44 between 2018 and 2025 (Stanford HAI AI Index 2026).

Major Milestones: March-May 2026

Date	Event	Significance
Mar 20, 2026	White House National AI Policy Framework	First federal AI legislative roadmap; preemption of state laws proposed
Apr 2, 2026	Microsoft releases MAI-Transcribe-1, MAI-Voice-1, MAI-Image-2	Microsoft builds independent model stack alongside OpenAI partnership
Apr 23, 2026	OpenAI releases GPT-5.5	Continued rapid iteration on frontier model capabilities
May 14, 2026	OpenAI launches Codex as standalone AI coding agent	Agentic coding goes mainstream; enterprise deployment via Dell partnership
May 19-20, 2026	Google I/O 2026: Gemini 3.5, Gemini Omni, Gemini Spark	Google positions Gemini as all-purpose AI hub; 900M+ monthly users
May 28, 2026	Anthropic raises \$65B at \$965B valuation; releases Claude Opus 4.8	Anthropic eclipses OpenAI in valuation; \$47B revenue run rate

3.1 FOCUS AREA: GENERATIVE AI

The past three months have seen an unprecedented pace of frontier model releases. OpenAI shipped GPT-5.5 (April 23), ChatGPT Images 2.0 (April 21), and Codex (May 14). Anthropic released Claude Opus 4.6, 4.7, and 4.8 in rapid succession. Google unveiled Gemini 3.5 Flash at I/O 2026. Microsoft released three MAI foundational models on April 2, 2026, at pricing designed to undercut Google and OpenAI.

Benchmark Leaders (May 2026)

Benchmark	Leader	Score	What It Measures
GPQA Diamond	GPT-5.4-Pro (OpenAI)	94.4%	Graduate-level science reasoning
SWE-Bench Verified	Claude Opus 4.7 (Anthropic)	87.6%	Real-world software engineering tasks

Implications for Technical Leaders

- Evaluate reasoning models (OpenAI o-series, Claude extended thinking) for complex engineering tasks where accuracy matters more than latency.
- Reassess context window requirements — 1M+ token contexts may eliminate chunking strategies in many RAG pipelines.
- Plan for multimodal integration in developer tooling: AI-assisted code review with diagram understanding, automated documentation with visual outputs.
- Microsoft MAI models offer cost-competitive alternatives for transcription (\$0.36/hr) and voice generation (\$22/million characters).

3.2 FOCUS AREA: AI INFRASTRUCTURE

AI infrastructure spending in 2026 has reached a scale without historical precedent. The four largest US hyperscalers have collectively raised their 2026 CapEx guidance to \$725B (Bloomberg, April 30, 2026). Including Chinese cloud providers and Oracle, the top nine global CSPs are projected to spend \$830B in 2026 — a 79% YoY increase (TrendForce, May 6, 2026). Gartner forecasts \$2.59T in total worldwide AI spending.

2026 Hyperscaler CapEx Commitments

Company	2026 CapEx	YoY Growth	Key Focus
Microsoft	\$190B	~130%	AI data centers, Azure AI, Copilot infrastructure
Amazon / AWS	\$230B+	>50%	AI cloud services, Trainium2, Inferentia
Google / Alphabet	\$180-190B	>100%	Gemini infrastructure, TPU v5p, data centers
Meta	\$125-145B	~85%	AI research, MTIA chips, Llama infrastructure
Top 9 CSPs Total	\$830B	79%	Global AI infrastructure buildout

AI Accelerator Market (2026)

- NVIDIA (~80% share):** B300 Blackwell Ultra — 1,800 TFLOPS FP16, 288GB HBM3e, ~\$350K+. GB300 NVL72 rack delivers 30x inference throughput vs. H100. CUDA ecosystem: 4M+ developers, 3,000+ optimized apps.
- AMD (~12-15% share):** MI325X — 288GB HBM3e, ~\$200K. ROCm 6.x now matches CUDA for most PyTorch/JAX workloads. Major wins at Meta, Microsoft, Oracle. 20-30% cheaper than equivalent NVIDIA systems.
- Custom Silicon:** Google TPU v5p (training), Amazon Trainium2 (40% cost savings vs. GPU on SageMaker), Microsoft Maia 100 (Azure Copilot), Meta MTIA (inference). Growing as hyperscalers optimize for specific workloads.
- Power & Cooling:** GB300 NVL72 rack draws ~120kW — requires purpose-built liquid cooling. AI servers projected to surpass general-purpose servers in total electricity consumption in 2026.

3.3 FOCUS AREA: AI AGENTS

Agentic AI has moved from emerging trend to operational reality. AI systems are no longer just answering questions — they are executing multi-step workflows autonomously, using tools, retaining context across sessions, and adapting based on outcomes.

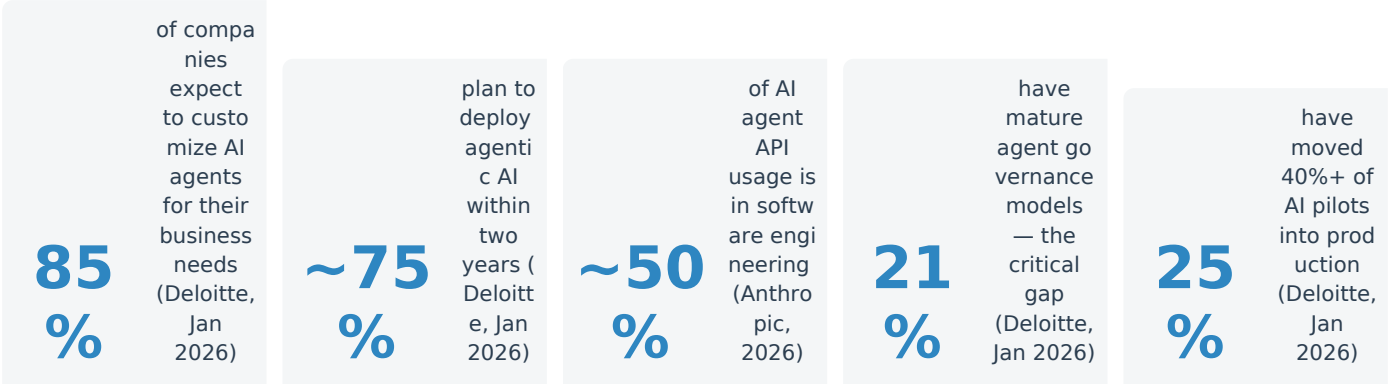
Key Commercial Agent Deployments (March-May 2026)

- OpenAI Codex (May 14, 2026):** Standalone AI coding agent, enterprise deployment via Dell Technologies for hybrid/on-premises environments. Competing directly with Anthropic's Claude Code.
- Anthropic Claude Code:** Annualized revenue of \$2.5B; business subscriptions quadrupled since start of 2026. Enterprise users represent >50% of revenue. Credited with driving ~\$2T selloff in traditional software stocks.

Google Gemini Spark (May 19, 2026): 24/7 personal AI agent operating across Gmail, Docs, Calendar. Cloud-based, no special hardware required. Creates custom workflows autonomously.

Google Gemini Omni: Multimodal AI agent generating video from text, image, and audio inputs. Rolling out to YouTube Shorts and Google Flow.

Enterprise Adoption Data



Leading Agent Frameworks

- LangGraph:** Best for complex, stateful workflows with branching logic. Open-source, LangChain ecosystem.
- CrewAI:** Best for role-based multi-agent teams and business process automation.
- Microsoft Semantic Kernel:** Best for enterprise AI and Microsoft 365/Azure ecosystem integration. Powers Copilot.
- AutoGen:** Best for multi-agent collaboration and conversational workflows.
- LlamaIndex:** Best for data-driven agents and knowledge retrieval systems (RAG).

3.4 FOCUS AREA: FOUNDATION MODELS

Foundation model competition has reached a new intensity, with major releases from every leading lab in the past three months. The competitive dynamics have shifted: it is no longer sufficient to have a capable model — labs must demonstrate enterprise reliability, coding performance, multimodal breadth, and cost efficiency simultaneously.

Frontier Model Landscape (May 2026)

Lab	Latest Model	Context	Key Strength	Valuation
OpenAI	GPT-5.5 (Apr 23, 2026)	196K+	Reasoning, GPQA Diamond leader (94.4%)	\$852B post-money
Anthropic	Claude Opus 4.8 (May 28, 2026)	200K+	Coding, SWE-Bench leader (87.6%), enterprise	\$965B (May 28)
Google DeepMind	Gemini 3.5 Flash (May 19, 2026)	1M-2M	Ultra-long context, multimodal, agentic	Public (Alphabet)
Microsoft MAI	MAI-Image-2 (Apr 2, 2026)	N/A	Cost-competitive transcription, voice, image/video	Public (Microsoft)
Meta	LLaMA 4 Enterprise (Dec 2025)	128K	Open-weight, enterprise fine-tuning, cost efficiency	Public (Meta)

Funding Milestones

- Anthropic: \$65B Series H at \$965B valuation (May 28, 2026) — eclipses OpenAI for first time. Revenue run rate: \$47B. Claude Code annualized revenue: \$2.5B. IPO preparation underway.
- OpenAI: \$122B funding round at \$852B post-money valuation (March 2026). IPO filing expected as soon as September 2026.
- Anthropic previously raised \$30B at \$380B valuation (Feb 12, 2026) — second-largest private tech financing round on record at the time.
- Scientific foundation models: Polymathic AI's Walrus and AION-1 demonstrate physics-trained models enabling cross-disciplinary scientific discovery (Jan 2026, Cambridge/Berkeley).

4. CROSS-CUTTING THEMES

1. Phase Transition: Experimentation to Execution

Organizations have moved from asking 'should we use AI?' to deploying it across entire workflows. 25% of leaders report AI is having a transformative effect — more than double from a year ago (Deloitte 2026). However, only 30% are redesigning key processes around AI, and 37% use AI at a surface level only.

2. Agentic AI as the New Architectural Paradigm

The shift from stateless model inference to persistent, tool-using agents represents the most significant architectural change in enterprise AI since RAG. Agents require fundamentally different infrastructure: persistent memory, tool registries, audit logging, and failure recovery.

3. Infrastructure as Geopolitical Strategy

The \$725-830B in hyperscaler CapEx is not just a technology investment — it is a geopolitical statement. AI sovereignty has emerged as a central policy principle. 77% of companies factor country of origin into vendor selection. US CHIPS Act and EU Chips Act are reshaping semiconductor supply chains.

4. The Governance Gap

Only 21% of companies deploying agentic AI have mature governance models. 68% of organizations have experienced data leakage from employees sharing sensitive information with AI tools. The International AI Safety Report 2026 (backed by 30+ countries) documents significant gaps in technical safeguards.

5. Energy as a Strategic Constraint

Data center electricity demand is projected to double by 2030 (Gartner). AI servers will surpass general-purpose servers in total electricity consumption in 2026. A single NVIDIA GB300 NVL72 rack draws ~120kW. Energy costs and availability are becoming strategic variables in AI infrastructure planning.

5. HARDWARE & INFRASTRUCTURE TRENDS

Note: No pre-existing AI Hardware Manufacturing Trends workflow document was available. Content sourced from live internet research.

The semiconductor industry has entered a sustained growth phase driven by AI demand, breaking from its historical boom-bust cycle. Data processing applications are on track to surpass 50% of total semiconductor revenue for the first time in 2026. High-Bandwidth Memory (HBM) is at the center of the investment wave.

Semiconductor supply chain: US CHIPS Act and EU Chips Act incentivizing new fab construction. TSMC N2 (2nm) entered volume production late 2025 — ~15% performance improvement, 25-30% power reduction vs. N3. Competition for TSMC wafer allocation is intense among Apple, NVIDIA, AMD, and Qualcomm.

AI accelerator market >\$150B: NVIDIA ~80% share (B300 at \$350K+, GB300 NVL72 at 30x H100 inference throughput). AMD 12-15% share (MI325X at ~\$200K, ROCm 6.x matches CUDA). Custom silicon growing at all major hyperscalers.

Power density crisis: Total installed data center power capacity expected to reach ~155 GW globally in 2026 (+29% YoY). AI servers projected to surpass general-purpose servers in total electricity consumption. Further jumps expected in 2027-2028 as GB300/Rubin and ASIC-based servers enter mass production.

Liquid cooling imperative: High-density AI compute clusters require advanced liquid cooling infrastructure. HVDC power systems and liquid cooling components are growth areas. Sustainability moving from corporate goal to regulatory mandate.

6. WORKFORCE & EMPLOYMENT IMPACT

Note: No pre-existing AI Employment Trends workflow document was available. Content sourced from live internet research.

Federal Reserve / Duke Study (Mar 25, 2026)

Survey of ~750 corporate executives:

- Widespread but uneven AI adoption
- Positive productivity gains, strengthening in 2026
- Limited near-term aggregate job loss
- Routine clerical roles declining; technical roles rising
- Productivity paradox: perceived > measured gains

Harvard Business School (Mar 4, 2026)

Analysis of nearly all US job postings 2019-Mar 2025:

- Automation-prone job postings: -13% post-ChatGPT
- Analytical/technical/creative roles: +20%
- AI is reshaping, not uniformly erasing, white-collar work
- BCG 2026: 'AI will reshape more jobs than it replaces'
- Human-AI collaboration is the dominant pattern

Skill Shifts for Technical Teams

In-demand skills: Prompt engineering, human-AI collaboration, domain-specific AI application development, AgentOps, AI governance and security.

Declining demand: Routine clerical coding tasks (boilerplate generation, basic documentation, simple test writing) are increasingly automated.

Reskilling imperative: Organizations must invest in reskilling programs. Retraining is essential for jobs where generative AI is reducing skill diversity (HBS, 2026).

Security literacy: 68% of organizations have experienced data leakage from employees sharing sensitive information with AI tools — AI security literacy is now a required competency.

7. STRATEGIC OUTLOOK & RECOMMENDATIONS

The next 6-12 months will be defined by three dynamics: the agentic AI buildout, the IPO wave, and regulatory crystallization. Technical leaders who act now on the recommendations below will be positioned to lead rather than react.

Forward-Looking Dynamics

The agentic AI buildout. ~75% of companies plan to deploy agentic AI within two years (Deloitte).

Organizations investing in AgentOps capabilities and multi-agent orchestration infrastructure in 2026 will have a significant competitive advantage. Demand for agent architecture expertise will intensify.

The IPO wave. OpenAI (targeting September 2026), Anthropic (preparing), and SpaceXAI (prospectus filed) are all moving toward public markets. Anticipate potential pricing changes and ensure AI vendor contracts include appropriate protections before IPOs change the commercial landscape.

Regulatory crystallization. The White House National AI Policy Framework (March 2026) signals the direction of federal AI regulation. Targeted legislation on child safety and online harms is most likely to advance near-term. The International AI Safety Report 2026 (30+ countries) is building international consensus on AI risk frameworks. Build compliance-ready AI architectures now.

5 Recommendations for Technical Leaders

1

Architect for agents, not just models

Redesign AI infrastructure to support persistent, tool-using agents as a first-class workload. Invest in AgentOps tooling (monitoring, logging, tool access controls, failure recovery) before scaling agent deployments. Evaluate LangGraph, Microsoft Semantic Kernel, or CrewAI based on your stack.

2

Close the governance gap before it closes you

Only 21% of companies have mature agent governance. Build governance infrastructure proactively: data access controls, audit logging, human-in-the-loop checkpoints for high-stakes decisions, and AI security training for all technical staff.

3

Adopt a multi-provider model strategy

No single foundation model leads across all dimensions. Maintain evaluation pipelines for Claude (coding, long-context), GPT-5.x (reasoning, professional workloads), and Gemini (multimodal, Google ecosystem). Benchmark continuously — the capability landscape shifts weekly.

4

Factor energy into AI infrastructure TCO

Power and cooling costs are becoming material variables. Evaluate liquid cooling requirements for high-density GPU deployments, assess energy efficiency of cloud vs. on-premises options, and build energy cost projections into multi-year AI infrastructure plans.

5

Invest in AI-native talent development

Prioritize reskilling programs focused on prompt engineering, human-AI collaboration, agent architecture, and AI governance. The organizations winning with AI invest in both their people and their tools simultaneously (Deloitte, 2026).

8. SOURCES & METHODOLOGY

All data points in this report are sourced from the publications listed below. No data has been fabricated or estimated. Where specific figures are cited, the source and date are noted in the text. No pre-existing workflow documents were available; all content is sourced from live internet research conducted on May 31, 2026.

Source	Type	Date	Sections
Deloitte State of AI in the Enterprise 2026	Survey (3,235 leaders)	Jan 2026	Agents, Workforce, Cross-cutting
Gartner AI Spending Forecast	Analyst report	Jan 15 & May 19, 2026	Infrastructure, Market size
TrendForce CSP CapEx Analysis	Market research	May 6, 2026	Infrastructure
Bloomberg: Big Tech CapEx	News	Apr 30 & Feb 5, 2026	Infrastructure
CNBC / Bloomberg: Anthropic funding	News	Feb 12 & May 28, 2026	Foundation Models
TechCrunch: Google I/O 2026	News	May 19, 2026	Generative AI, Agents
TechCrunch: Microsoft MAI models	News	Apr 2, 2026	Foundation Models
OpenAI Newsroom	Official announcements	May 2026	Generative AI
AI Release Tracker	Model tracking database	May 2026	Generative AI, Foundation Models
Stanford HAI AI Index 2026	Research report	2026	Regulation, Policy
International AI Safety Report 2026	International research	Feb 3, 2026	Regulation, Safety
White House National AI Policy Framework	Government document	Mar 20, 2026	Regulation
Federal Reserve Atlanta Working Paper 2026-4	Academic research	Mar 25, 2026	Workforce
Harvard Business Review / HBS	Academic research	Mar 4, 2026	Workforce

BCG 2026 AI Workforce Report	Consulting report	2026	Workforce
Data-Gate.ch GPU Market Analysis	Industry analysis	May 31, 2026	Hardware
Intertek Semiconductor Blog	Industry analysis	Feb 17, 2026	Hardware
AdvancedAI.com AI Landscape 2026	Industry analysis	Mar 28, 2026	Cross-cutting
Anthropic 2026 State of AI Agents Report	Industry report	2026	Agents
Turing.com AI Agent Frameworks Guide	Technical guide	Feb 11, 2026	Agents

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